

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech Branch : Civil Engineering /Civil and Infrastructure Engineering

Semester : IV

Subject Code & Name: BTCVC402/BTCIC402 Environmental Engineering

Max Marks: 60

Date: 21/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Objective type questions. (Compulsory Question)

12

- 1 Which one of the following methods gives the best estimate of population growth of a community with limited land area for future expansion?
a. Arithmetical increase method b. Geometrical increase method c. Incremental increase method d. Logistic method
- 2 If the average daily water consumption of a city is 24000 cum, the peak hourly demand (of the maximum day) will be
a. 1000 cum/h b. 1500 cum/hr c. 1800 cum/hr d. 2700 cum/hr
- 3 The threshold odour number (TON) for a water sample of 40 ml, diluted to standard 200 ml mixture, in which odour is just barely detectable to the sense of smell, is
a. 8 b. 5 c. 50 d. None of these
- 4 Which of the following chemical compounds can be used for dechlorination of water?
a. carbon dioxide b. bleaching powder c. sulphur dioxide d. chloramines
- 5 The maximum permissible limit For fluoride in drinking water is
a. 0.1 mg/l b. 1.5 mg/l c. 5 mg/l d. 10 mg/l
- 6 The valve, which allows the flow only in one direction, is a
a. reflux valve b. sluice valve c. gate valve d. None of these

7	Double filtration is used		1
	a. to increase the filtration capacity of slow sand filters	b. to increase the filtration capacity of rapid sand filters	c. for isolated colonies of a few houses
	d. for none of the above		
8	The formula, which is most appropriate for the design of pressure pipes is		1
	a. Darcy Weisbach formula	b. Mannings formula	c. Chezy's formula
	d. Dupuit's formula		
	Which one of the following solid waste disposal methods is ecologically most acceptable?		1
	a. Sanitary landfill	b. Incineration	c. Composting
	d. Pyrolysis		
10	Allowable disposable rate of application of sludge on land is determined by		1
	a. carbon content of sludge	b. nitrogen content of sludge	c. phosphorus content of sludge
	d. potassium content of sludge		
	Symbiosis, the beneficial association between algae and bacteria is used for treatment of waste water in the following unit?		1
	a. Activated Sludge	b. Rotating Biological Disc	c. Anaerobic Digester
	d. Oxidation Pond		
	The relative stability of a sewage sample, whose D.O. equals the total oxygen required to satisfy its BOD, is		1
	a. zero	b. 1%	c. 100%
	d. infinity		
Q. 2 Solve the following.			12
A)	Explain various chemical characteristics of water? What are the various methods of determining the chemical characteristics?	BT2/CO2	6
B)	What are various processes required to remove the various types of impurities? Give a neat sketch showing the various processes required in treating a river water, starting from the source of water.	BT2/CO3	6
Q. 3 Solve the following.			12
A)	What are the requirements of good distribution systems? Describe in brief various types of distribution systems.	BT1/CO2	6
B)	What are the various chemical coagulants which are commonly used in coagulation process? How they remove suspended impurities?	BT2/CO2	6

Q. 4 Solve Any Two of the following.		12
A) Write a short note on	BT3/CO2	6
a. Break point chlorination		
b. Free available chlorine		
c. Chlorine demand		
B) Explain various physical and Chemical characteristics of wastewater.	BT2/CO2	6
C) Explain clearly how does a rapid gravity filter differ in its action from a slow sand filter?	BT3/CO3	6
Q. 5 Solve Any Two of the following.		12
A) Explain Primary & Secondary treatment of waste water in detail.	BT2/CO1	6
B) Differentiate between BOD and COD.	BT2/CO3	6
C) In a BOD test, 1.0 ml of raw sewage was diluted to 100 ml and the dissolved oxygen concentration of diluted sample at the beginning was 6 ppm and it was 4 ppm at the end of 5-day incubation at 20°C. Find out the BOD of raw sewage.	BT4/CO3	6
Q. 6 Solve Any Two of the following.		12
A) What are the harmful effects of polluted air on human beings?	BT2/CO4	6
B) Define solid waste. Why it is necessary to manage solid waste. Explain various classification of solid waste based on source generation and characteristics.	BT2/CO4	6
C) Explain the following atmospheric condition:	BT2/CO4	6
a. Super adiabatic and Sub adiabatic		
b. Neutral & Inversion		
How do they influence the dispersion of pollutants in the atmosphere?		
*** End ***		

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